

Assignment 9: Time Series Analysis

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A09_TimeSeries.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Check your working directory
 - Load the tidyverse, zoo, and trend packages
 - Set your ggplot theme

```
library(tidyverse)
library(lubridate)
library(trend)
library(zoo)
library(here)

here()
```

```
## [1] "/home/guest/EDE_Spring2026_personal"
```

```
mytheme <- theme_minimal(base_size = 12) +
  theme(axis.text = element_text(color = "navy"), legend.position = "top",
        plot.title = element_text(face = "bold"))
theme_set(mytheme)
```

2. Import the ten datasets from the `Ozone_TimeSeries` folder in the Raw data folder. These contain ozone concentrations at Garinger High School in North Carolina from 2010-2019 (the EPA air database only allows downloads for one year at a time). Import these either individually or in bulk and then combine them into a single dataframe named `GaringerOzone` of 3589 observation and 20 variables.

```

#1
EPAair_2019 <- read.csv(
  here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2019_raw.csv"),
  stringsAsFactors = TRUE)
EPAair_2018 <- read.csv(
  here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2018_raw.csv"),
  stringsAsFactors = TRUE)
EPAair_2017 <- read.csv(
  here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2017_raw.csv"),
  stringsAsFactors = TRUE)
EPAair_2016 <- read.csv(
  here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2016_raw.csv"),
  stringsAsFactors = TRUE)
EPAair_2015 <- read.csv(
  here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2015_raw.csv"),
  stringsAsFactors = TRUE)
EPAair_2014 <- read.csv(
  here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2014_raw.csv"),
  stringsAsFactors = TRUE)
EPAair_2013 <- read.csv(
  here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2013_raw.csv"),
  stringsAsFactors = TRUE)
EPAair_2012 <- read.csv(
  here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2012_raw.csv"),
  stringsAsFactors = TRUE)
EPAair_2011 <- read.csv(
  here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2011_raw.csv"),
  stringsAsFactors = TRUE)
EPAair_2010 <- read.csv(
  here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2010_raw.csv"),
  stringsAsFactors = TRUE)

GaringerOzone <- rbind(
  EPAair_2010, EPAair_2011, EPAair_2012, EPAair_2013, EPAair_2014,
  EPAair_2015, EPAair_2016, EPAair_2017, EPAair_2018, EPAair_2019
)

dim(GaringerOzone)

```

```
## [1] 3589 20
```

Wrangle

3. Set your date column as a date class.
4. Wrangle your dataset so that it only contains the columns Date, Daily.Max.8.hour.Ozone.Concentration, and DAILY_AQI_VALUE.
5. Notice there are a few days in each year that are missing ozone concentrations. We want to generate a daily dataset, so we will need to fill in any missing days with NA. Create a new data frame that contains a sequence of dates from 2010-01-01 to 2019-12-31 (hint: `as.data.frame(seq())`). Call this new data frame Days. Rename the column name in Days to "Date".

6. Use a `left_join` to combine the data frames. Specify the correct order of data frames within this function so that the final dimensions are 3652 rows and 3 columns. Call your combined data frame `GaringerOzone`.

```
# 3
GaringerOzone$Date <- as.Date(GaringerOzone$Date, format = "%m/%d/%Y")

# 4
GaringerOzone_wrangle <- GaringerOzone %>%
  select(Date, Daily.Max.8.hour.Ozone.Concentration, DAILY_AQI_VALUE)

# 5
Days <- as.data.frame(seq(
  from = as.Date("2010-01-01"),
  to   = as.Date("2019-12-31"),
  by   = "day"
))

colnames(Days) <- "Date"

# 6
GaringerOzone <- Days %>% left_join(GaringerOzone_wrangle, by = "Date")
```

Visualize

7. Create a line plot depicting ozone concentrations over time. In this case, we will plot actual concentrations in ppm, not AQI values. Format your axes accordingly. Add a smoothed line showing any linear trend of your data. Does your plot suggest a trend in ozone concentration over time?

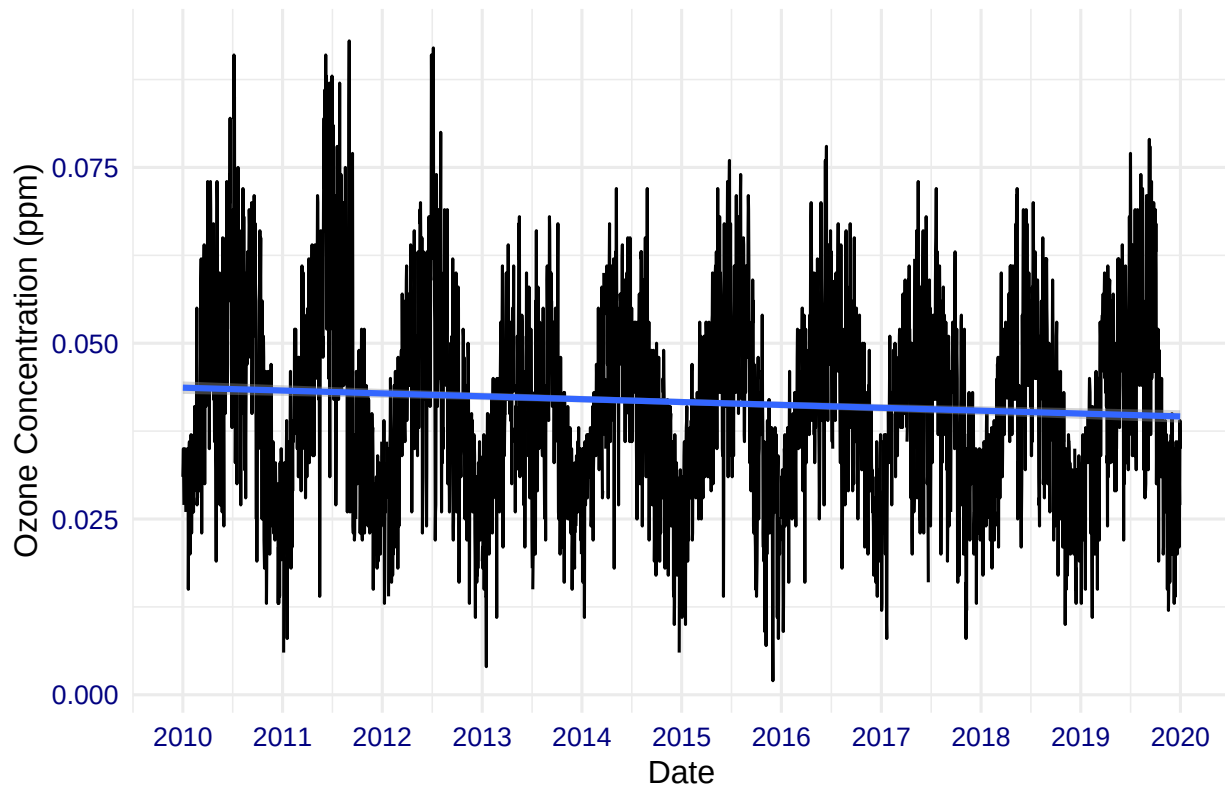
```
#7
Ozone_over_time <-
  ggplot(GaringerOzone, aes(x=Date, y=Daily.Max.8.hour.Ozone.Concentration)) +
  geom_line() +
  geom_smooth(method = lm) +
  scale_x_date(
    date_breaks = "1 year", date_labels = "%Y") +
  labs(
    title = "Daily Ozone Concentrations over Time (2010-2019)",
    x = "Date",
    y = "Ozone Concentration (ppm)")

print(Ozone_over_time)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 63 rows containing non-finite outside the scale range
## ('stat_smooth()').
```

Daily Ozone Concentrations over Time (2010–2019)



Answer: The smoothed line suggests an overall decreasing trend in daily ozone concentrations, although this is a slight decrease. The plot itself shows that ozone concentrations might be cyclical because they go up and down throughout each year.

Time Series Analysis

Study question: Have ozone concentrations changed over the 2010s at this station?

8. Use a linear interpolation to fill in missing daily data for ozone concentration. Why didn't we use a piecewise constant or spline interpolation?

#8

#checking for NAs - there are NAs

`summary(GaringerOzone)`

```
##      Date      Daily.Max.8.hour.Ozone.Concentration  DAILY_AQI_VALUE
##  Min.   :2010-01-01      Min.   :0.00200              Min.    : 2.00
## 1st Qu.:2012-07-01      1st Qu.:0.03200              1st Qu.: 30.00
## Median :2014-12-31      Median :0.04100              Median  : 38.00
## Mean   :2014-12-31      Mean   :0.04163              Mean    : 41.57
## 3rd Qu.:2017-07-01      3rd Qu.:0.05100              3rd Qu.: 47.00
## Max.   :2019-12-31      Max.   :0.09300              Max.    :169.00
##      NA's      NA's      :63              NA's    :63
```

```

GaringerOzone_clean <-
  GaringerOzone %>%
  mutate(Ozone_interp =
    zoo::na.approx(Daily.Max.8.hour.Ozone.Concentration))

#checking for NAs - they are now gone!
summary(GaringerOzone_clean$Ozone_interp)

```

```

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00200 0.03200 0.04100 0.04151 0.05100 0.09300

```

Answer: The gaps in ozone data are very short, so connecting the dots between values using a linear interpolation is the best method for this. Piecewise constant interpolation wouldn't work because it assumes that missing values are to be equal to the measurement made nearest to that date. This is likely to be incorrect for ozone data because it varies daily due to different variables. Spline interpolation could work because it uses a quadratic function to interpolate instead of drawing a straight line, but this might exaggerate variability in the data.

9. Create a new data frame called `GaringerOzone.monthly` that contains aggregated data: mean ozone concentrations for each month. In your pipe, you will need to first add columns for year and month to form the groupings. In a separate line of code, create a new Date column with each month-year combination being set as the first day of the month (this is for graphing purposes only)

```

#9
GaringerOzone.monthly <- GaringerOzone_clean %>%
  mutate(
    Year = year(Date),
    Month = month(Date)) %>%
  group_by(Year, Month) %>%
  summarize(
    Mean_Ozone = mean(Ozone_interp)
  ) %>%
  select(Year, Month, Mean_Ozone)

```

```

## 'summarise()' has regrouped the output.
## i Summaries were computed grouped by Year and Month.
## i Output is grouped by Year.
## i Use 'summarise(.groups = "drop_last")' to silence this message.
## i Use 'summarise(.by = c(Year, Month))' for per-operation grouping
##   ('?dplyr::dplyr_by') instead.

```

```

GaringerOzone.monthly <-
  GaringerOzone.monthly %>%
  mutate(Date = as.Date(paste0(Year, "-", Month, "-01")))

```

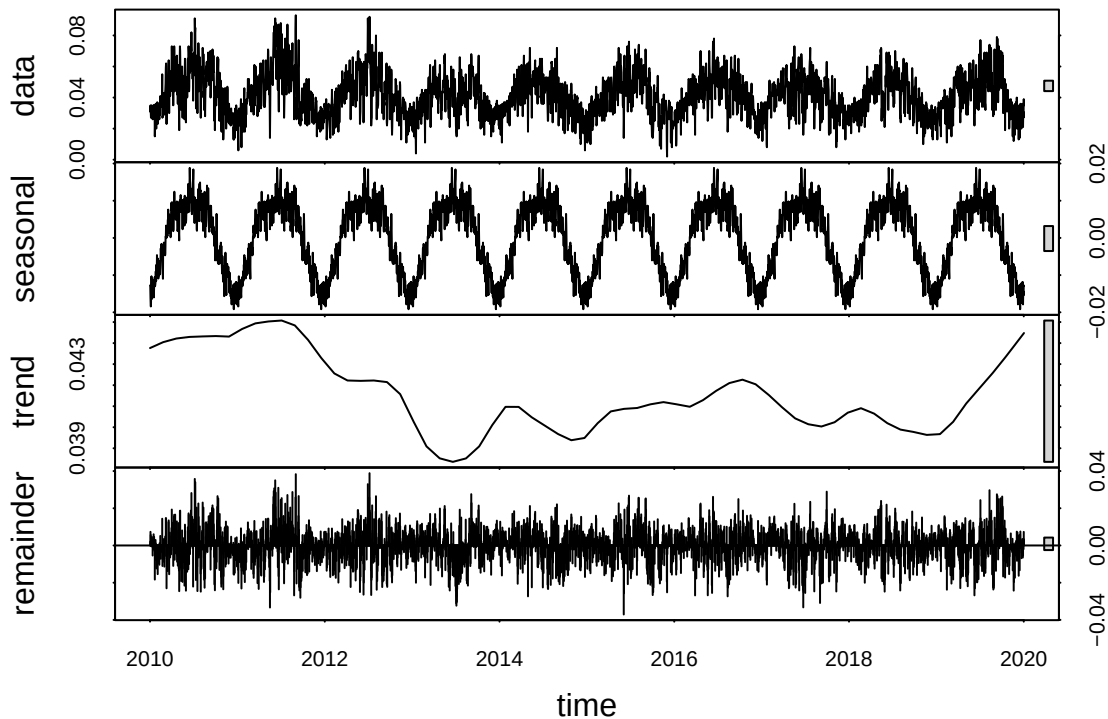
10. Generate two time series objects. Name the first `GaringerOzone.daily.ts` and base it on the dataframe of daily observations. Name the second `GaringerOzone.monthly.ts` and base it on the monthly average ozone values. Be sure that each specifies the correct start and end dates and the frequency of the time series.

```
#10
f_year <- year(first(GaringerOzone_clean$Date))
GaringerOzone.daily.ts <- ts(GaringerOzone_clean$Ozone_interp,
                             start=c(f_year,1),
                             frequency = 365)

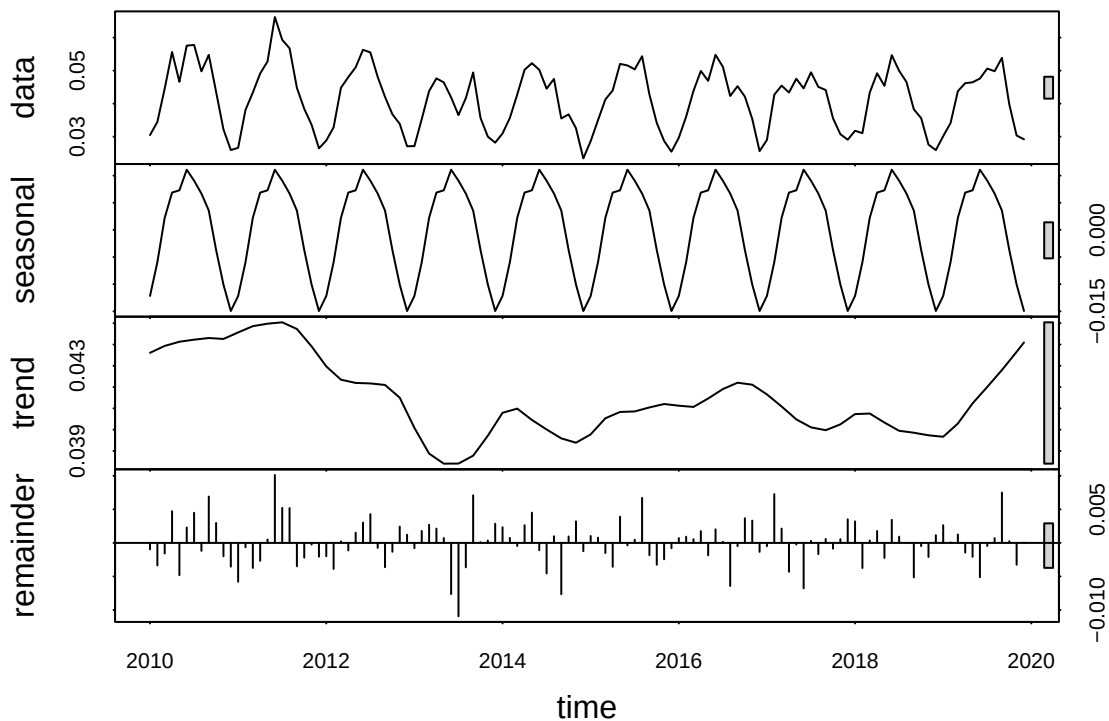
f_month <- month(first(GaringerOzone.monthly$Date))
f_year <- year(first(GaringerOzone.monthly$Date))
GaringerOzone.monthly.ts <- ts(GaringerOzone.monthly$Mean_Ozone,
                                start=c(f_year, f_month),
                                frequency=12)
```

11. Decompose the daily and the monthly time series objects and plot the components using the `plot()` function.

```
#11
Garinger_daily_decomp <- stl(GaringerOzone.daily.ts,
                             s.window="periodic")
plot(Garinger_daily_decomp)
```



```
Garinger_monthly_decomp <- stl(GaringerOzone.monthly.ts,
                                s.window="periodic")
plot(Garinger_monthly_decomp)
```



12. Run a monotonic trend analysis for the monthly Ozone series. In this case the seasonal Mann-Kendall is most appropriate; why is this?

```
#12
GaringerOzone_monthly_trend <-
  Kendall::SeasonalMannKendall(GaringerOzone.monthly.ts)

GaringerOzone_monthly_trend
```

```
## tau = -0.143, 2-sided pvalue =0.046724
```

```
summary(GaringerOzone_monthly_trend)
```

```
## Score = -77 , Var(Score) = 1499
## denominator = 539.4972
## tau = -0.143, 2-sided pvalue =0.046724
```

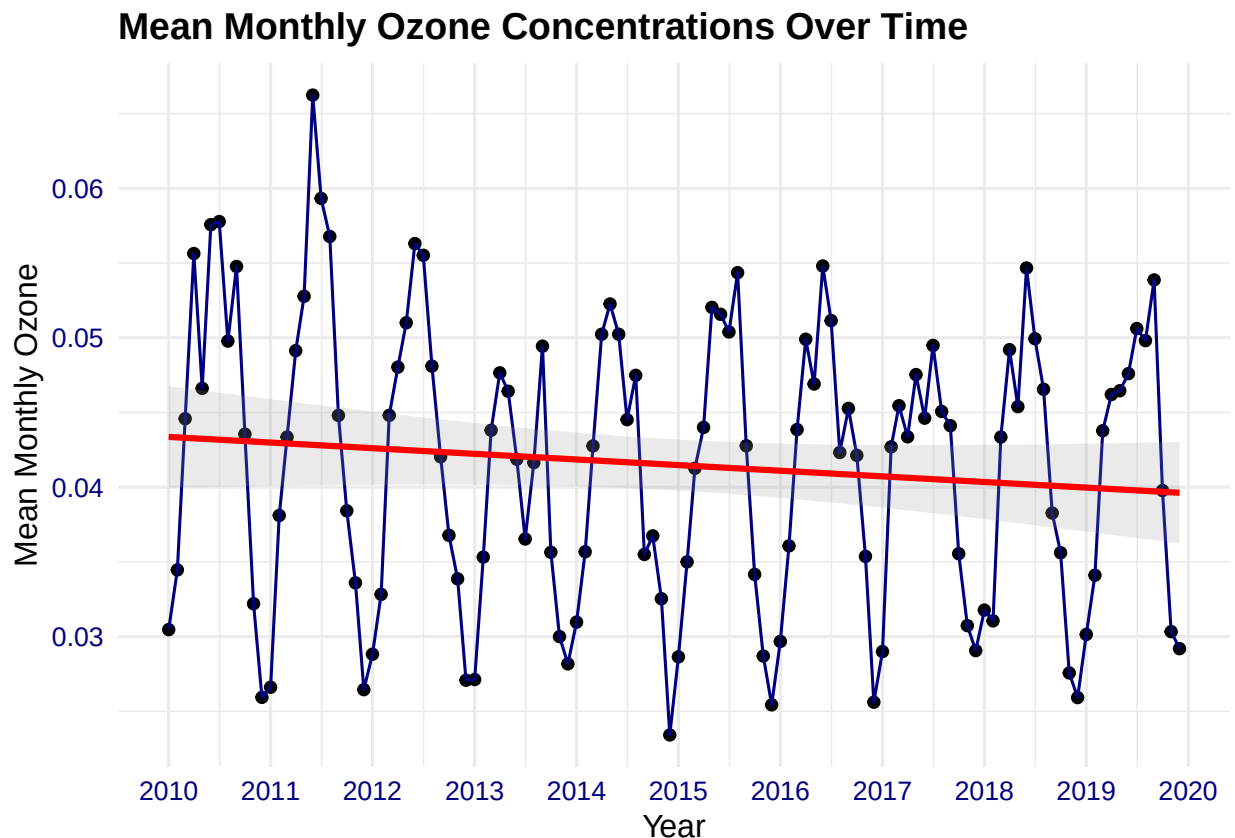
Answer: This monotonic trend analysis that the $\tau = -0.143$ means there is a slight decrease over time in monthly mean ozone concentrations. The p value is 0.0467 which is less than 0.05, so there is statistically significant evidence of a trend over time. This type of test is the best for this data because it is strongly seasonal. Linear regression, Mann-Kendall, and Spearman Rho wouldn't work because they can only be used for data that is not seasonal.

13. Create a plot depicting mean monthly ozone concentrations over time, with both a `geom_point` and a `geom_line` layer. Edit your axis labels accordingly.

```
# 13
Ozone_monthly_plot <-
  ggplot(GaringerOzone.monthly, aes(x=Date, y=Mean_Ozone)) +
  geom_point() +
  geom_line(color="navy") +
  geom_smooth(method = lm, color = "red", alpha=0.2) +
  scale_x_date(
    date_breaks = "1 year", date_labels = "%Y") +
  labs(y="Mean Monthly Ozone", x="Year",
    title="Mean Monthly Ozone Concentrations Over Time")

print(Ozone_monthly_plot)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



14. To accompany your graph, summarize your results in context of the research question. Include output from the statistical test in parentheses at the end of your sentence. Feel free to use multiple sentences in your interpretation.

Answer: The decomposition plot from #11 shows clear seasonal variation in mean monthly ozone concentrations, supporting my use of a seasonal Mann-Kendall test for trend analysis. The results show a weak negative trend, suggesting that monthly mean ozone concentrations have slightly decreased over time. This trend is statistically significant, as the p-value is less than 0.05, indicating evidence of a monotonic decline. This finding is reflected in the time series plot

from #13, which shows a slight downward trend in ozone concentrations over the study period ($\tau = -0.143$, $p = 0.0467$).

15. Subtract the seasonal component from the `GaringerOzone.monthly.ts`. Hint: Look at how we extracted the series components for the `EnoDischarge` on the lesson Rmd file.
16. Run the Mann Kendall test on the non-seasonal Ozone monthly series. Compare the results with the ones obtained with the Seasonal Mann Kendall on the complete series.

```
#15
GaringerOzone_monthly_nonseas <-
  GaringerOzone.monthly.ts -
  Garinger_monthly_decomp$time.series[,1]

#16
Ozone_nonseas_trend <-
  trend::mk.test(GaringerOzone_monthly_nonseas)
Ozone_nonseas_trend

##
## Mann-Kendall trend test
##
## data: GaringerOzone_monthly_nonseas
## z = -2.672, n = 120, p-value = 0.00754
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##          S          varS          tau
## -1.179000e+03  1.943657e+05 -1.651376e-01
```

Answer: The results from this test are fairly similar from the seasonal test. The non-seasonal Mann-Kendall test on monthly ozone concentrations also showed a negative trend ($\tau = -0.165$, $p = 0.0075$), which means a decrease in ozone over time. Compared to the seasonal Mann-Kendall test ($\tau = -0.143$, $p = 0.0467$), the non-seasonal test shows a slightly stronger trend and smaller p-value. By ignoring the seasonal variability, the statistical significance increases and this might be an overestimation.